Assignment 3

Due date: TBA.
Total: 30 points
Coverage: Chapter 4: Generating Random Variables and Chapter 5: Monte Carlo Simulation and Variance Reduction.

Question	Format	Points
1	8 multiple-choice questions on generators and normalization	8
2	8 multiple-choice questions on simulation and variance reduction	8
3	6 true/false questions on interpreting simulation	6
4	One shop example with 8 short, structured responses	8
Total		**30**

Assignment PDF | Response template (.qmd)

Submission

Use **R Markdown** or **Quarto**. Submit both your **source file** (`.Rmd` or `.qmd`) and its **rendered PDF** to the **Assignment 3** folder on **iCollege**. Use the filename `A3_LastName_FirstName` with the appropriate extension and include your name in the PDF.
Begin with the answer table for Questions 1-3, followed by the eight labeled responses for Question 4. Use the supplied response template, or follow the same labels and order.

- **Questions 1-2:** Write one letter, A, B, C, or D, for each item.
- **Question 3:** Write **True** or **False** for each item.
- **Question 4:** Follow each prompt's requested format. A short sentence or calculation is enough; no essays are needed.

Every item is worth **1 point**. No explanations are required for Questions 1-3. **No code execution or plots are required.** If you use R to check an answer, include the code in your source file. For random sampling or simulation, set a seed (e.g., `set.seed(8670)`) before generating random values and include it in your submitted code so your results are reproducible. Your source must render to the submitted PDF without errors.

Scope: Use the main material in Chapters 4 and 5. **Supplementary material is not assessed.** Focus on what the simulation does, what it estimates, and why its output varies. No proofs or derivations of optimal variance-reduction coefficients are required.

Question 1: Generators and Normalization [8]

Select the **best answer**. Each item is worth **1 point**.

1a)

Consider the following R code:

```r
x <- numeric(100)
for (i in seq_len(100)) {
  set.seed(8670)
  x[i] <- runif(1)
}
```

What is the main problem with using `x` as 100 independent uniform draws?
A. The values must all be zero.
B. The values are sampled without replacement.
C. Resetting the seed repeats the same first draw in every iteration.
D. A uniform generator cannot be used inside a loop.

1b)

Let $U \sim \text{Uniform}(0, 1)$. Which expression generates an exponential random variable with **mean 5**?
A. $-\log(1-U)/5$
B. $-5\log(1-U)$
C. $\exp(-5U)$
D. $5U$

1c)

The Beta$(2, 2)$ density is $f(x) = 6x(1-x)$ for $0 < x < 1$, and zero otherwise. Its height at $x = 0.5$ is 1.5. Which statement is correct?
A. It cannot be a density because its height exceeds 1.
B. $P(X = 0.5) = 1.5$.
C. The area under the curve is 1.5.
D. It is a valid density: its total area is 1, even though its height can exceed 1.

1d)

An unnormalized target is $h(x) = x(1-x)$ on $(0, 1)$, with total area $1/6$. Which formula turns it into a normalized density?
A. $f(x) = 6h(x)$
B. $f(x) = h(x)/6$
C. $f(x) = h(x) - 1/6$
D. $f(x) = h(x)$

1e)

For the same target, compare two rejection samplers with a uniform proposal on $(0, 1)$:

- normalized scale: target $f(y) = 6y(1 - y)$, envelope height 1.5;
- unnormalized scale: target $h(y) = y(1 - y)$, envelope height 0.25.

Using the same proposal Y and decision draw $U \sim \text{Uniform}(0, 1)$, independent of Y, how do the acceptance decisions compare?
A. The normalized version accepts six times as often.
B. The unnormalized version cannot be used for rejection sampling.
C. Both accept exactly when $U \leq 4Y(1 - Y)$.
D. Both accept exactly when $U \leq Y(1 - Y)$.

1f)

Use the normalized sampler in 1e). For one proposal, $Y = 0.75$ and $U = 0.60$. Its acceptance ratio is $f(Y)/1.5 = 0.75$. What should the sampler do?
A. Accept and store 0.75.
B. Accept and store 0.60.
C. Reject because $U < 0.75$.
D. Accept and store the density height 1.125.

1g)

In the rejection diagram, a proposal is plotted at (Y, V) with $V = 1.5U$, where U is uniform on $(0, 1)$ and independent of Y. What does the point's height V represent?
A. The probability that the target equals Y.
B. The value of the target density $f(Y)$.
C. A second accepted draw from the target distribution.
D. An auxiliary random height between zero and the envelope, used to decide whether to accept Y.

1h)

A programmer chooses an envelope after checking the target at only 100 grid points. What must be established for a valid rejection sampler? Assume the proposal covers the target support.
A. The number of proposals is at least 100.
B. The envelope is at least as high as the target throughout its support.
C. The envelope has total area exactly 1.
D. Every proposal is accepted.

Question 2: Simulation and Variance Reduction [8]

Select the **best answer**. Each item is worth **1 point**.

2a)

Why use simulation to estimate the chance of an event when a direct calculation is difficult?
A. It guarantees that the chosen model matches the real world.
B. It removes all uncertainty after a single run.
C. It generates repeated outcomes from a model and estimates how often the event occurs.
D. It replaces the event probability by the mean of the model's inputs.

2b)

A shop has 12 muffins each day. The vector `demand` contains the number of customers wanting one muffin on each simulated day. Which R expression estimates the probability that **at least one customer cannot be served**?
A. `mean(demand > 12)`
B. `mean(demand)`
C. `sum(demand > 12)`
D. `mean(demand >= 12)`

2c)

Two runs use the same model and the same number of simulated days, but fresh random draws. Their estimated shortage probabilities differ. What is the main reason?
A. The stock level must have changed.
B. The model's true shortage probability changes after each run.
C. One of the runs must contain a coding error.
D. Different finite samples contain different numbers of shortage days.

2d)

A simulation averages independent outputs $Y_1, \ldots, Y_N$. What do the output SD and the MCSE describe?
A. Both measure only the spread of individual Y_i values.
B. SD describes the spread of individual outputs; MCSE describes the uncertainty of their average due to simulation.
C. SD describes the uncertainty of the average; MCSE describes individual outputs.
D. MCSE measures how wrong the model's assumptions are.

2e)

With 1,000 independent simulated days, an estimated probability has MCSE about 0.02. With the same model and method, what MCSE should we expect using 4,000 days?
A. About 0.01.
B. About 0.005.
C. About 0.02.
D. Exactly zero.

2f)

What is the goal of variance reduction in this chapter?
A. Change the quantity of interest to one with a smaller numerical value.
B. Make real customer demand less variable by simulating more days.
C. Estimate the same quantity with less run-to-run variation at comparable computational work.
D. Select the random seed that happens to give the closest answer.

2g)

An antithetic method generates 500 independent $U_j \sim \text{Uniform}(0,1)$ draws and forms the 500 pair averages $A_j = \{e^{U_j} + e^{1-U_j}\}/2$. Which values are the independent units used in $\text{sd}(\cdot)/\sqrt{N}$ to estimate MCSE?
A. The 1,000 separate function values, treating all of them as independent.
B. The 500 pair averages, so MCSE is $\text{sd}(A)/\sqrt{500}$.
C. Only the first pair average.
D. The two values in any single pair.

2h)

For $U \sim \text{Uniform}(0,1)$, a control-variate method uses $Y_{\text{adj}} = e^U - c(U - 0.5)$, with a fixed coefficient c. Why does this adjustment preserve the target mean $\mathbb{E}[e^U]$?
A. The adjusted output equals e^U for every draw.
B. Any constant can replace 0.5 without changing the mean.
C. The adjustment always makes the output positive.
D. The subtracted term has mean zero because $\mathbb{E}[U] = 0.5$.

Question 3: Interpreting Simulation [6]

Write **True** or **False** for each statement. Each item is worth **1 point**. No explanation is required.

3a)

Setting the seed once at the start allows the same simulation to be reproduced when the code and computing environment are unchanged.

3b)

Increasing the number of simulated days makes real customer demand less variable from day to day.

3c)

In rejection sampling, increasing a valid envelope by a constant factor greater than 1 preserves the distribution of accepted values but lowers the acceptance rate.

3d)

Pairing U and $1 - U$ must reduce Monte Carlo variance for every possible function of U.

3e)

A small MCSE guarantees that the model assumptions accurately describe the real world.

3f)

To compare simulation methods, we should use repeated runs targeting the same quantity and compare their variability at comparable computational work.

Question 4: From One Artificial Day to Many Runs [8]

Each numbered prompt is worth **1 point**. Give only the requested sentence, expression, or calculation. Use the labels **4a.1-4a.4** and **4b.1-4b.4** in your response.

4a) Why simulate, and what are the steps? [4]

A shop prepares **12 muffins** each morning. Assume **20 potential customers** per day. Each independently wants one muffin with probability **0.6**, whether or not stock is available. Days are independent. These are illustrative model assumptions, not estimates from shop data.

Let D be the number of customers wanting a muffin in one day. We want to estimate $p = P(D > 12)$, the probability that at least one customer cannot be served.

1. The mean demand is $20 \times 0.6 = 12$. Why does this **not** imply that 12 muffins are always enough? Give **one sentence**.
2. Describe how to generate **one artificial day's demand** D using independent uniform draws. State how many draws you use and which draws count as customers wanting a muffin. Give **one sentence**; R code is also acceptable.
3. Complete the rule: record $Y = 1$ if _____; otherwise record $Y = 0$. Give an **inequality involving** D.
4. In one run of 1,000 independent days, 430 days have insufficient stock. Compute $\hat{p}$. Give the **fraction and its decimal value**.

4b) Why do repeated runs differ? [4]

A second run of the same model, using fresh random draws for another 1,000 days, has 397 shortage days. Stock, purchase probability, and all other assumptions are unchanged.

1. The two estimates are 0.430 and 0.397. Explain why they can differ even if both programs are correct. Give **one sentence**.
2. For the first run, use $\hat{p} = 0.430$ and $N = 1{,}000$ to compute $\widehat{\text{MCSE}} = \sqrt{\hat{p}(1-\hat{p})/N}$. Give the value to **four decimal places**.
3. If N increases to 4,000 and $\hat{p}$ stays close to 0.430, what approximate MCSE do you expect? Give the value to **four decimal places**.
4. Complete both statements using **decreases**, **stays the same**, or **increases**: when only the number of simulated days increases, the typical run-to-run spread of $\hat{p}$ _____; the model's day-to-day spread of demand D _____.